

# Chapter 166 — Management of Castration-Resistant Prostate Cancer

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- **Suppression of gonadal testosterone** is the central principle of androgen deprivation therapy (ADT) — one of the most effective systemic treatments known for any solid tumour.
- **Progression despite castrate testosterone is universal:** almost every patient on ADT eventually shows biochemical and clinical progression, reflecting adaptive survival mechanisms in the tumour.
- **Historical outcome with ADT alone** (orchiectomy or luteinising hormone-releasing hormone [LHRH]-targeted therapy) in metastatic disease: progression-free survival (PFS) 12–20 months, overall survival (OS) 24–36 months; conventional endocrine therapy alone improved only modestly over decades.
- **Life-prolonging classes now available** in castration-resistant prostate cancer (CRPC): next-generation androgen receptor (AR)-pathway inhibitors (ARPIs); taxane chemotherapy; radiopharmaceuticals (radium-223 and prostate-specific membrane antigen [PSMA]-targeted lutetium-177); poly-adenosine diphosphate (ADP)-ribose polymerase inhibitors (PARPIs); and immunotherapy.
- **Castration-resistant clones exist early:** cells with the castration-resistant phenotype are identifiable in the early stages of prostate cancer.
- **The AR stays central:** somatic AR alterations are frequent after ADT; a molecularly altered AR can be activated by other ligands (oestrogens, progestational agents) and ligand-independently by growth factors and cytokines — the rationale for ARPIs such as abiraterone and enzalutamide.
- **Proliferation–apoptosis balance:** with androgens present, proliferation exceeds cell death; ablation triggers a swift apoptotic cascade; as the tumour progresses the apoptotic threshold rises until proliferation again exceeds death, and endocrine-independent cells come to dominate.
- **Lineage plasticity** (neuroendocrine/anaplastic differentiation) is common under treatment, ranging from bone-predominant disease to aggressive variants with visceral spread and low PSA; these may need platinum regimens, as in small cell lung carcinoma.
- **Other targets:** peptide growth factors and cell-surface receptors such as PSMA (targets for radiopharmaceuticals, antibody–drug conjugates and monoclonal antibodies); **BRCA1/BRCA2** mutations create synthetic lethality with PARPIs (olaparib, talazoparib, rucaparib).

**Concept — castration resistance is not androgen independence:** the AR remains the principal driver, so the chapter's therapeutic logic runs from deeper AR suppression first to non-AR mechanisms (DNA repair, bone, PSMA, immunity) once AR dependence is lost.

## CLINICAL CONSIDERATIONS

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### Disease States

- **TNM (tumour, node, metastasis) staging fails in advanced disease:** purely anatomic, it does not guide management of recurrent or advanced prostate cancer.
- **Clinical states model** (Prostate Cancer Clinical Trials Working Group, 2008; updated as PCWG3, 2016) classifies patients by response to prior therapy; the chapter's prognostic and therapeutic reasoning is built on it.
- **Baseline evaluation:** extent of disease; mode and site of progression (PSA rise alone, new bone metastasis, visceral or nodal metastasis); symptoms, especially bone pain; response to previous endocrine therapy.

**FIG. 166.1 PROSTATE CANCER CLINICAL STATES MODEL (PCWG3, SCHER ET AL., 2016)**

**Clinically localised disease → rising PSA (non-castrate) → clinical metastases (non-castrate) or non-metastatic CRPC (nmCRPC) → metastatic CRPC (mCRPC) first line → second line → third, fourth ... line X.**  
 The first three states are non-castrate; nmCRPC and every mCRPC line are castration-resistant.

**Monitoring progression**

- **PSA, computed tomography/nuclear medicine (CT/NM) imaging and symptoms** remain the main tools for assessing progression.
- **PSA doubling time (PSADT) ≤10 months** defined high-risk nmCRPC in SPARTAN, ARAMIS and PROSPER, the pivotal trials that set the nmCRPC standard of care.
- **Rising PSA is usually the first sign** of progression; historically, with ADT alone, it preceded bone-scan progression while patients stayed largely asymptomatic.
- **Radiographic progression without PSA progression is now common** because ARPIs are used earlier: in ARCHES (enzalutamide + ADT in hormone-sensitive disease) up to two-thirds of radiographic progressors had no concurrent PSA progression — **PSA alone is insufficient** for these patients.
- **Check serum testosterone** when non-compliance is suspected or prior regimens do not sustain castration (non-steroidal antiandrogen monotherapy, low-dose oestrogens, 5α-reductase inhibitors).

**Continuing ADT**

- **Maintain LHRH agonist/antagonist indefinitely**, including during chemotherapy and ARPIs; almost all CRPC trials mandate continued testosterone suppression (medical or surgical).
- **Rationale:** stopping suppression in non-orchietomised men may worsen progression and survival; exogenous testosterone can cause a severe flare (pain, neurological, urological and coagulation complications) in a minority. Some studies show no detriment to stopping, but consensus is to continue.

**Antiandrogen withdrawal**

- **Withdrawal of a first-generation antiandrogen** (e.g. bicalutamide; steroidal or non-steroidal) can give short-lived PSA falls, symptomatic benefit and occasionally objective responses in a minority.
- **First step when progressing on combined androgen blockade:** observed discontinuation of the antiandrogen → serial PSA for **4–8 weeks** → next manoeuvre.

**Choosing the next therapy**

- **Options:** second-line hormonal manipulation (first-generation antiandrogen or ARPI), cytotoxic chemotherapy, radiopharmaceuticals or molecularly targeted therapy.
- **Second-line hormonal therapy is most common first;** bicalutamide lowers PSA with little toxicity, but ARPIs have largely replaced first-generation agents over the last decade.
- **Sequential hormonal approach** suits limited, asymptomatic metastatic disease (e.g. rising PSA without pain).
- **ARPI + PARPI** should be considered in patients progressing on ADT who carry homologous recombination repair (HRR) mutations, especially BRCA.
- **Sequencing is unknown:** the mCRPC trials enrolled men progressing on ADT alone, whereas hormone-sensitive disease is now treated upfront with chemotherapy and ARPIs (abiraterone, enzalutamide, apalutamide; darolutamide with docetaxel).

**Predicting behaviour**

- **PSADT predicts** the speed of bone-scan progression and survival; **PSADT <3 months** → particularly rapid course → consider more aggressive management.

- **Symptomatic bone pain or visceral-predominant disease** → upfront cytotoxic chemotherapy may serve better.
- **Poorly differentiated and neuroendocrine tumours** seldom respond durably to further AR-directed therapy.
- **Biopsy of metastatic sites** in clinically aggressive disease with relatively low PSA may reveal a neuroendocrine phenotype on immunostaining; routine biopsy of all such patients is not yet defined.

### Genomic testing

- **mCRPC genomes differ from the primary**; the current standard for somatic next-generation sequencing is tumour tissue, usually the untreated primary, because metastatic (mostly bone) biopsy is difficult.
- **Liquid biopsy** (circulating tumour DNA) is an alternative with limits: **clonal haematopoiesis of indeterminate potential (CHIP)** can produce spurious DNA-repair-gene calls — such patients are unlikely to respond to PARPIs despite a "positive" report.
- **Germline testing is recommended** for: metastatic prostate cancer; family history of BRCA1/2-, Lynch syndrome- or HOXB13-associated cancers; Ashkenazi ancestry.

**Concept** — *classify by clinical state, not anatomy: prior therapy, pace (PSADT) and phenotype decide what comes next, and because ARPIs now come earlier, the old PSA-first pattern of progression can no longer be relied upon.*

## NONMETASTATIC CASTRATION-RESISTANT PROSTATE CANCER

- **Stage migration created the M0 CRPC group**: ADT started at first PSA rise after local therapy, before metastases are visible on conventional imaging. The group grew but has likely peaked as **PSMA PET** reclassifies many as metastatic.
- **PSA kinetics are the strongest predictor** of metastatic progression (PSA velocity, PSADT); pre-ADT factors (pretreatment PSA, PSADT, stage, Gleason score) and response to ADT also contribute.
- **Natural history can be long** in unselected men with rising PSA on ADT: about one-third have bone metastases at 2 years; median time to bone metastasis about 30 months. **Baseline PSA >10 ng/mL** and **PSA velocity** independently predict time to bone metastasis and survival.
- **Old approach**: sequential first-generation agents (bicalutamide, ketoconazole). **Current standard**: an ARPI when **PSADT ≤10 months**.
- **Three ARPIs are approved for nmCRPC** — apalutamide (first; SPARTAN), enzalutamide (PROSPER) and darolutamide (ARAMIS), each improving metastasis-free survival (MFS) and OS (trial details under Androgen Signaling Inhibitors).
- **Unresolved**: how PSMA PET will redefine this population and whether treatment should follow hormone-sensitive or castration-resistant logic.

**Concept** — *nmCRPC is an imaging artefact defined by kinetics: it exists because conventional scans miss disease, so PSADT selects who needs treatment and more sensitive imaging is shrinking the category.*

## METASTATIC CASTRATION-RESISTANT PROSTATE CANCER

- **Heterogeneous population**; trial eligibility and risk stratification still rest on conventional imaging (bone scintigraphy and CT).
- **Most mCRPC remains addicted to AR signalling** through AR overexpression, enhancer and gene amplification, and transcriptionally active AR splice variants lacking the ligand-binding domain.

### Bone tropism

- **Overwhelming predilection for bone:** circulating cells arrest in cortical and medullary bone → adhere via receptors for integrins, collagens, laminin and other bone proteins → grow under hormones, growth factors and stromal-epithelial interactions in the marrow.
- **Consequences:** pain, nerve-root or cord compression, pathological fracture; marrow replacement → **myelophthitic anaemia and thrombocytopenia**.
- **Visceral metastasis in <20%**, soft-tissue nodal disease in 30–40%; visceral disease without bone involvement is rarer still.
- **Response rate is a poor end point:** soft-tissue shrinkage reflects a small fraction of total (mostly bone) burden → PCWG3 prefers **PFS incorporating bone and soft-tissue radiographic progression**.

### Haematological complications

- **Anaemia is the commonest:** chronic disease, marrow invasion, blood loss, rarely microangiopathic haemolysis with disseminated intravascular coagulation; compounded by pelvic marrow irradiation, chemotherapy, radiopharmaceuticals and long-term ADT.
- **Erythropoiesis-stimulating agents:** not recommended, or only with caution — they may increase mortality in cancer patients.
- **Thrombocytopenia** (rarely leukopenia) usually follows extensive radiotherapy or chemotherapy; **pancytopenia** is an end-stage marrow process.
- **Thrombocytosis** is non-specific; clotting complications are rare and treatment is usually unnecessary.

### Urological and neurological emergencies

- **Obstructive uropathy** (primary tumour or pelvic/retroperitoneal nodes) → infection, pain and renal impairment that can preclude renally cleared chemotherapy → **relieve obstruction first** (internal ureteric stent or percutaneous nephrostomy) in chemotherapy candidates.
- **Spinal cord compression** is among the greatest oncological emergencies; vertebral involvement makes it common and it needs expeditious management (see Palliative Management).

**Concept — bone dominance shapes everything:** it produces the complications (pain, fracture, cord compression, marrow failure) and explains why response rate misleads and PFS is preferred.

## EVALUATION OF TREATMENT EFFICACY

- **Consistent prognostic factors** across settings: performance status, pain, haemoglobin, PSA, alkaline phosphatase, lactate dehydrogenase (LDH), extent of bone involvement (number or pattern of lesions) and visceral disease.

### RESPONSE ASSESSMENT BY MODALITY

| Modality                        | Behaviour on treatment                                                                                                                                                            | Rule / interpretation                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| PSA                             | Declines predict outcome on abiraterone and enzalutamide; chemotherapy may cause a rise before a fall; <b>little PSA change</b> in the sipuleucel-T and radium-223 pivotal trials | <b>PSA flare/fluctuation</b> on radium-223 and $^{177}\text{Lu}$ -PSMA-617 does not imply worse prognosis |
| Circulating tumour cells (CTCs) | Falling counts associate with better outcome on chemotherapy and on abiraterone/enzalutamide                                                                                      | Prognostic <b>independently of PSA</b> ; still investigational                                            |

| Modality    | Behaviour on treatment                                                                                                                | Rule / interpretation                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Bone scan   | Rarely shows response; sclerosis may mean healing or progression; new lesions may be a <b>flare</b> of subclinical responding disease | PCWG3: <b>new baseline scan at 12–16 weeks</b> ; thereafter progression = <b>≥2 new lesions</b> ; clinical improvement or PSA fall supports flare |
| CT          | Measurable or evaluable non-bone disease                                                                                              | <b>RECIST</b> (Response Evaluation Criteria in Solid Tumours)                                                                                     |
| PSMA PET/CT | Apparent progression within 4 weeks resolving by 3 months (~13%, later responders)                                                    | Insufficient data in CRPC; response frameworks still being developed                                                                              |

- **What the table tests:** in CRPC every single marker can mislead early — PSA, bone scan and PSMA PET all flare — so progression is called on confirmed, combined evidence, with the 12–16-week scan as the reference.

**Concept — early apparent progression is often flare; the PCWG3 new-baseline and "two new lesions" rules exist to stop effective therapy being abandoned too soon.**

#### KEY POINTS: CLINICAL CONSIDERATIONS

- **Evaluate extent and aggressiveness** of disease before choosing therapy.
- **Critical determinants:** radiographic metastases on conventional imaging or not; biochemical vs radiographic vs clinical progression; symptoms (e.g. pain).
- **mCRPC is heterogeneous** in metastatic distribution, disease kinetics and treatment response.

## NEUROENDOCRINE/AGGRESSIVE VARIANT METASTATIC PROSTATE CANCER

- **Neuroendocrine/anaplastic transformation** occurs in a small proportion of advanced cases; it is **inherently endocrine-resistant** and needs different treatment.
- **Biology shared with neuroendocrine tumours elsewhere** (most often lung): receptors for neuroendocrine peptides (somatostatin, chromogranin A, serotonin), **parathyroid hormone-related protein (PTHrP)**, TP53 mutation, **low PSMA expression**.
- **Molecular signature** (small cell neuroendocrine prostate cancer): **at least two of three** — PTEN loss, TP53 mutation, RB1 loss.
- **Clinical pattern:** subacute, dramatic change — rapidly growing soft-tissue masses (primary site, retroperitoneum), liver infiltration, **osteolytic** bone metastases, parenchymal brain metastases.
- **PSA stops rising or becomes undetectable** despite major progression. PSMA PET is thought to be of low utility, though some tumours remain PSMA-avid and might benefit from PSMA-based treatment.
- **Biopsy is strongly encouraged** → small cell or poorly differentiated neoplasm with neuroendocrine immunostaining.
- **Treatment as for small cell lung cancer:** **cisplatin + etoposide** or **taxane + carboplatin**; doxorubicin-containing regimens modestly effective; clinical trials encouraged.
- **Radiotherapy** for bulky disease, brain metastases, or local control in critical sites (pain, impending fracture, bladder outlet obstruction); combined chemoradiotherapy is often needed.
- **Prognosis poor** despite high initial response: survival generally **<12 months**.

**USUAL ADENOCARCINOMA VS NEUROENDOCRINE/AGGRESSIVE VARIANT CRPC**

| Feature          | Usual mCRPC (adenocarcinoma)             | Neuroendocrine/aggressive variant                                   |
|------------------|------------------------------------------|---------------------------------------------------------------------|
| AR dependence    | Retained (AR-driven)                     | <b>Lost</b> — unresponsive to hormonal therapy                      |
| PSA              | Tracks disease                           | <b>Low, falling or undetectable</b> despite progression             |
| Bone lesions     | Osteoblastic                             | <b>Osteolytic</b> , with hypercalcaemia (PTHrP)                     |
| Metastatic sites | Bone ± nodes; visceral <20%              | <b>Visceral (liver), bulky pelvic/retroperitoneal masses, brain</b> |
| PSMA             | Usually expressed                        | <b>Low</b>                                                          |
| Treatment        | ARPI, taxane, radiopharmaceutical, PARPI | <b>Platinum-based chemotherapy</b> ± radiotherapy                   |
| Survival         | Years across sequential lines            | <b>&lt;12 months</b>                                                |

**Concept — the trap is a falling PSA with a growing tumour:** loss of AR dependence removes both the marker and the target, so rapid progression with low PSA should prompt biopsy and a switch to platinum.

**KEY POINTS: NEUROENDOCRINE PROSTATE CANCER**

- **Suspect the neuroendocrine subtype** in rapidly growing disease with pelvic masses, visceral involvement, osteolytic metastases with hypercalcaemia (high serum PTHrP) or parenchymal brain metastases.
- **PSA usually undetectable** (or low/declining) despite rapid progression; serum chromogranin A and urinary serotonin metabolites may be detectable but are not needed for diagnosis.
- **Invariably hormone-unresponsive**, but transiently sensitive to radiotherapy and chemotherapy (platinum–etoposide or platinum–taxane).

**ANDROGEN SIGNALING INHIBITORS****CYP17 Inhibition: Abiraterone and Other Agents**

- **AR signalling stays active** despite castrate testosterone (<50 ng/dL).
- **Residual androgen sources:** LHRH agents do not touch **adrenal androgens (up to 10% of total)**; CRPC may synthesise intratumoural androgens; CYP17 (cytochrome P450 17) is overexpressed in CRPC.
- **Pathway:** CYP17 converts precursors → dehydroepiandrosterone (DHEA) and androstenedione → testosterone → **dihydrotestosterone** (the most potent androgen). Hence **CRPC is not androgen independent**.
- **Abiraterone acetate:** oral selective CYP17 inhibitor; active after ketoconazole (a weak CYP17 inhibitor).

**Abiraterone: practical prescribing**

- **Dose:** 1000 mg orally daily with prednisone (10 mg daily in mCRPC trials) and continued LHRH agonist/antagonist.
- **Why prednisone:** CYP17 blockade reduces cortisol → ACTH drive → mineralocorticoid excess → **hypokalaemia, hypertension, pedal oedema**; prednisone blunts the ACTH drive.
- **Other toxicity:** raised liver enzymes (case reports of liver injury and jaundice in practice) → interruption, dose change or discontinuation; **increased mortality in pre-existing cardiovascular disease**.

- **FDA cautions:** cardiovascular disease, adrenocortical insufficiency, liver disease; control hypertension; **correct hypokalaemia before starting.**
- **Take on an empty stomach** — food increases bioavailability up to **tenfold**; alternative: 250 mg daily with a low-fat meal ( $\leq 5$  g fat,  $\sim 300$  kcal). A reformulation (Yonsa) allows more flexible dosing with equivalent efficacy.

### Other CYP17-pathway agents

- **Orteronel (TAK-700):** non-steroidal CYP17 inhibitor, more 17,20-lyase-selective; two phase III mCRPC trials (chemotherapy-naïve and post-docetaxel) and an mCSPC trial did not improve survival → no longer pursued.
- **ODM-208:** targets CYP11A1; given with steroids; phase I activity.
- **Abiraterone (and Yonsa) are the only FDA-approved CYP17 inhibitors.**
- **Moving earlier:** LATITUDE and STAMPEDE showed survival gains with ADT + abiraterone/prednisone in high-risk metastatic hormone-sensitive disease; in nmCRPC (IMAAGEN, abiraterone + prednisone 5 mg) PSA response  $\geq 50\%$  was met, but **abiraterone is not approved for nmCRPC**. Earlier use limits later utility in CRPC.

**Concept** — *abiraterone treats the androgen the testis does not make; every toxicity (mineralocorticoid excess) and every co-prescription (prednisone, continued ADT) follows from blocking CYP17 upstream of both cortisol and androgens.*

### Androgen Receptor Modulation: Enzalutamide, Apalutamide, Darolutamide and Other Agents

- **Resistance mechanisms:** constitutively active AR mutations, AR overexpression, bypass through accessory pathways (AKT/mTOR/PI3K, HER2/neu).
- **Next-generation antiandrogens gave survival gains** that first-generation agents (bicalutamide, nilutamide, flutamide) never did.
- **Enzalutamide:** potent non-steroidal AR antagonist with **no measurable agonist activity**, even with overexpressed or activated AR; also blocks **nuclear translocation** and **DNA binding** at androgen-response elements.
- **No steroid needed with enzalutamide** — it may be less active with prednisone (prednisone may activate a promiscuous AR or act via the glucocorticoid receptor).
- **Apalutamide (ARN-509):** pure AR antagonist that also inhibits nuclear translocation and DNA binding; **does not cross the blood–brain barrier**, no seizure signal; notable for **rash** and **hypothyroidism**.
- **Darolutamide:** blocks androgen binding; **structurally distinct**; does not cross the blood–brain barrier → less fatigue; **twice-daily** dosing.
- **Class toxicity:** fatigue is the most significant; **falls and fractures** → caution in the elderly ( $>75$  years).

### NEXT-GENERATION AR-PATHWAY INHIBITORS: COMPARISON

| Agent               | Target / mechanism                                                     | Dose                   | Steroid                    | Signature toxicity                                               | CRPC approvals                              |
|---------------------|------------------------------------------------------------------------|------------------------|----------------------------|------------------------------------------------------------------|---------------------------------------------|
| <b>Abiraterone</b>  | <b>CYP17 inhibitor</b><br>(androgen synthesis: adrenal, intratumoural) | 1000 mg daily, fasting | <b>Yes</b><br>(prednisone) | Hypokalaemia, hypertension, oedema, hepatotoxicity; CV mortality | mCRPC (pre- and post-docetaxel); not nmCRPC |
| <b>Enzalutamide</b> | AR antagonist + blocks translocation and DNA binding                   | 160 mg daily           | No                         | <b>Seizures (~1%)</b> , fatigue, falls                           | mCRPC (pre- and post-docetaxel); nmCRPC     |

| Agent               | Target / mechanism                                   | Dose                      | Steroid | Signature toxicity                                              | CRPC approvals                |
|---------------------|------------------------------------------------------|---------------------------|---------|-----------------------------------------------------------------|-------------------------------|
| <b>Apalutamide</b>  | AR antagonist + blocks translocation and DNA binding | 240 mg daily              | No      | <b>Rash, hypothyroidism, fatigue, falls; no BBB penetration</b> | nmCRPC (also mCSPC)           |
| <b>Darolutamide</b> | AR antagonist; distinct structure                    | <b>300 mg twice daily</b> | No      | Least CNS effect; no BBB penetration                            | nmCRPC (mCSPC with docetaxel) |

- **What the table is for:** only abiraterone needs a steroid; only enzalutamide carries the seizure warning; the two BBB-sparing agents (apalutamide, darolutamide) are the ones chosen to limit CNS toxicity; only enzalutamide holds both mCRPC and nmCRPC approvals.

- **Trap:** abiraterone is active in nmCRPC (IMAAGEN) but is **not approved** there.

#### PIVOTAL ARPI TRIALS IN CRPC

| Trial             | Population                                                   | Comparator                                 | Primary end point → result                                                                                | Verdict                                                     |
|-------------------|--------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>COU-AA-301</b> | mCRPC, post-docetaxel, ketoconazole-naïve; 2:1               | Placebo + prednisone                       | OS <b>14.8 vs 10.9 mo</b> , HR <b>0.65</b> ; rPFS 5.6 vs 3.6; PSA response 38% vs 10%                     | Pain, fatigue and skeletal-event benefit; mainstay of mCRPC |
| <b>COU-AA-302</b> | Chemotherapy-naïve, asymptomatic/mildly symptomatic mCRPC    | Placebo + prednisone                       | Co-primary rPFS and OS: rPFS significant; strong OS trend; chemotherapy delayed <b>26.5 vs 16.8 mo</b>    | Label extended to all mCRPC                                 |
| <b>AFFIRM</b>     | mCRPC after 1–2 chemotherapy lines (incl. docetaxel)         | Placebo                                    | OS <b>18.4 vs 13.6 mo</b> , HR <b>0.63</b> ; PSA ≥50% fall 54% vs 1.5%; time to progression 8.3 vs 3.0 mo | Post-docetaxel approval (enzalutamide 160 mg)               |
| <b>PREVAIL</b>    | Chemotherapy-naïve, asymptomatic/minimally symptomatic mCRPC | Placebo                                    | Death risk ↓ 29% (HR <b>0.71</b> ); radiographic progression ↓ 81% (HR <b>0.19</b> )                      | Label extended to all mCRPC                                 |
| <b>PROSPER</b>    | nmCRPC, PSADT <10 mo                                         | Placebo (+ ADT)                            | MFS improved; death risk ↓ 27%                                                                            | Enzalutamide approved for nmCRPC                            |
| <b>SPARTAN</b>    | nmCRPC, PSADT ≤10 mo                                         | Placebo                                    | MFS: risk ↓ 72% (HR <b>0.28</b> ); OS <b>73.9 vs 59.9 mo</b> , HR <b>0.78</b>                             | Apalutamide: first nmCRPC drug                              |
| <b>ARAMIS</b>     | nmCRPC, PSADT ≤10 mo                                         | Placebo                                    | MFS <b>40.4 vs 18.4 mo</b> ; death HR 0.69; delayed symptomatic skeletal events and chemotherapy          | Darolutamide approved for nmCRPC; no safety signal          |
| <b>IMAAGEN</b>    | High-risk nmCRPC                                             | Single arm (abiraterone + prednisone 5 mg) | PSA fall ≥50% achieved                                                                                    | Not approved for nmCRPC                                     |

- **Pattern:** each ARPI was proven first after docetaxel, then before it, then in nmCRPC (and mCSPC) — the drugs migrated forward, which is why most men now reach CRPC already exposed to one.
- **Apalutamide in mCSPC (TITAN):** longer OS and time to chemotherapy vs placebo; rash showed the largest exposure-adjusted excess, otherwise a similar safety profile.

## How New Approvals in the Metastatic Castration-Sensitive Space Affect Treatment for mCRPC

- **mCSPC standards now:** ADT + docetaxel; + abiraterone/prednisone; + enzalutamide; + apalutamide; + docetaxel + darolutamide; + docetaxel + abiraterone/prednisone. Fewer men reach mCRPC on ADT alone.
- **Cross-resistance:** progression on one ARPI predicts resistance to another; sequential ARPI benefits only 15–30% — try only with close monitoring.

### NEXT OPTION AT MCRPC BY PRIOR MCSPC THERAPY

| Prior therapy           | Reasonable next options                                              |
|-------------------------|----------------------------------------------------------------------|
| Docetaxel (no ARPI)     | <b>ARPI</b>                                                          |
| ARPI (no chemotherapy)  | <b>Chemotherapy</b> or non-hormonal agent (radium-223, sipuleucel-T) |
| Both docetaxel and ARPI | <b>Radiopharmaceutical</b> or <b>cabazitaxel</b>                     |

- **Practical selection factors:** comorbidity (infection, diabetes, accelerated cardiovascular disease, liver disease, seizure disorder); co-medications; cost (oral co-payments may make chemotherapy the realistic option); dosing burden (darolutamide's twice-daily schedule).
- **Drug-drug interactions** (anticoagulants, antiretrovirals, antidepressants especially):
  - **Enzalutamide:** avoid **CYP2C8 inhibitors** (raise its exposure); strong/moderate CYP3A4 or CYP2C8 **inducers** lower its exposure; it **reduces exposure** of narrow-therapeutic-index drugs metabolised by CYP2C9, CYP2C19 or CYP3A4.
  - **Abiraterone:** metabolised by CYP3A4 (inhibitors raise exposure); avoid co-prescribed CYP2D6 substrates.

**Concept — prior exposure now dictates the CRPC menu:** ARPI-on-ARPI rarely works, so the logic is to switch mechanism — AR ↔ taxane ↔ radioligand — rather than intensify the same one.

#### KEY POINTS: ANDROGEN SIGNALING INHIBITORS

- **Next-generation ARPIs improve OS** in castration-sensitive disease (abiraterone, enzalutamide, apalutamide; darolutamide with docetaxel) and in CRPC (abiraterone, enzalutamide).
- **Cross-resistance** limits ARPIs used in sequence.
- **Three drugs approved for nmCRPC:** apalutamide, enzalutamide, darolutamide.

## NOVEL ANDROGEN RECEPTOR-TARGETING APPROACHES

### Androgen Receptor Degraders

- **Rationale:** resistant mCRPC shows AR overexpression, increased nuclear translocation, and ligand-binding-domain mutations or splice variants that stabilise AR by reducing proteolysis → **degrade the receptor** instead of blocking it.
- **PROTAC (proteolysis-targeting chimera):** tags a target protein for destruction by the ubiquitin-proteasome system; highly selective, fewer off-target effects, can reach "undruggable" targets.
- **ARV-110:** phase I in heavily pretreated mCRPC (≥2 prior therapies incl. enzalutamide or abiraterone); manageable toxicity; PSA falls ≥50% (PSA50) in **46% of those with AR T878A/S and/or H875Y mutations**; later trials stratify by AR mutation.

- **ARV-766**: degrades wild-type AR and ligand-binding-domain mutants including the commonest, **L702H**; further PROTACs are in preclinical development.

**Concept — *degradation defeats the mutations that defeat antagonists***: when the receptor is mutated or overabundant, removing the protein works where competitive blockade fails.

## CYTOTOXIC CHEMOTHERAPY

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### Chemotherapeutic Agents

- **Historically tried**: cyclophosphamide, 5-fluorouracil, estramustine, vinorelbine, etoposide, cisplatin, carboplatin, doxorubicin, mitoxantrone, paclitaxel, docetaxel.
- **Still in use**: docetaxel, cabazitaxel, occasionally platinum, perhaps mitoxantrone — the rest gave neither symptom nor survival benefit.
- **Survival on chemotherapy** in advanced CRPC: about 16–20 months.

### Docetaxel

- **Mechanism**:  $\beta$ -tubulin binding  $\rightarrow$  microtubule stabilisation (GTP-independent polymerisation)  $\rightarrow$  **G2/M arrest**; TP53-independent apoptosis; BCL2 phosphorylation (caspase-3 activation; loss of anti-apoptotic function; BAX no longer inhibited); CDKN1B (p27) induction; BCL-XL repression; **impairs microtubule transport of ligand-bound AR** into the nucleus.
- **2004: drug of choice for mCRPC** (TAX 327), superseding mitoxantrone + prednisone.

### TAX 327 (landmark)

- **Design**: 1006 chemotherapy-naïve men with stable pain; all on prednisone 5 mg twice daily and continued ADT; three arms — mitoxantrone 12 mg/m<sup>2</sup> 3-weekly; **docetaxel 75 mg/m<sup>2</sup> 3-weekly**; docetaxel 30 mg/m<sup>2</sup> weekly; planned 30 weeks (10 cycles).
- **Result**: OS **18.9 months** (3-weekly docetaxel) vs 17.3 (weekly) vs 16.4 (mitoxantrone); 3-weekly docetaxel cut death risk by **24% (HR 0.76, 95% CI 0.62–0.94, P = 0.009)**; pain and PSA responses also best with docetaxel.
- **First demonstration that chemotherapy prolongs survival in mCRPC.**
- **Schedule contrast**: 3-weekly  $\rightarrow$  more neutropenia and febrile neutropenia, slightly more neuropathy; weekly  $\rightarrow$  more nausea/vomiting, fatigue, nail change, hyperlacrimation, diarrhoea. Quality of life (FACT-P) similar between schedules, better than mitoxantrone.

### SWOG 9916

- **Docetaxel 60 mg/m<sup>2</sup> + estramustine** vs mitoxantrone + prednisone: OS **17.5 vs 15.6 months, HR 0.80**.
- **Estramustine abandoned**: thromboembolism (required warfarin and aspirin prophylaxis), gastrointestinal and cardiovascular toxicity, and no added activity over docetaxel alone.

### Failed docetaxel combinations

- **No OS gain** from adding anti-angiogenics (bevacizumab, aflibercept, lenalidomide — despite VEGF correlating inversely with survival), bone-targeted agents (atrasentan, zibotentan, dasatinib) or high-dose calcitriol.
- **Why they failed**: marginal partner activity; no randomised phase II before phase III; docetaxel dose reductions from overlapping toxicity.

### Docetaxel moved upfront

- **Docetaxel + ADT** improves survival in hormone-sensitive disease; **triplets** (ADT + docetaxel + ARPI) beat ADT + docetaxel — **PEACE-1** (abiraterone) and **ARASENS** (darolutamide).

- **At CRPC:** re-treat with docetaxel, or move to cabazitaxel if progression occurred on docetaxel.

## Cabazitaxel

- **Second-generation taxane** with **poor affinity for P-glycoprotein** (ATP-dependent efflux pump) → active in docetaxel-sensitive and docetaxel-resistant tumours; neutropenia was dose-limiting in phase I.
- **TROPIC (landmark; FDA approval June 2010):** 755 men progressing after docetaxel; cabazitaxel 25 mg/m<sup>2</sup> 3-weekly + prednisone vs mitoxantrone 12 mg/m<sup>2</sup> + prednisone → OS **15.1 vs 12.7 months (HR 0.70)**; better PFS, PSA and radiological responses; **no difference in pain response**.
- **Subgroups:** benefit regardless of measurable disease, pain, or progression on vs after docetaxel; greatest with ECOG (Eastern Cooperative Oncology Group) 0–1 and in those progressing **<3 months** from docetaxel start → little cross-resistance with docetaxel.
- **Toxicity:** grade 3–4 **neutropenia 82%** (febrile 8%), worse >65 years; diarrhoea (worse >65 and after prior radiotherapy); fatigue; **neuropathy rarely severe** (grade 3 ~1%). Growth factor support recommended.
- **PROSELICA:** 20 mg/m<sup>2</sup> **non-inferior** to 25 mg/m<sup>2</sup> for OS with fewer grade 3–4 events (40% vs 55%) → label changed to **20 mg/m<sup>2</sup> for most**; 25 mg/m<sup>2</sup> reasonable if low neutropenic risk.
- **FIRSTANA:** cabazitaxel (20 or 25 mg/m<sup>2</sup>) **not superior to docetaxel first line** → cabazitaxel is **not** approved as first-line chemotherapy.
- **CARD:** after docetaxel and one ARPI (abiraterone or enzalutamide), **cabazitaxel beat a second ARPI** — OS **13.6 vs 11.0 months (HR 0.64)**; PFS 4.4 vs 2.7 months (HR 0.52).

## Mitoxantrone

- **First chemotherapy approved for CRPC (FDA 1997, with prednisone)**, for symptomatic disease: semisynthetic anthracycline; best palliation with low-dose corticosteroid.
- **Two randomised trials** (vs prednisone alone; vs hydrocortisone alone) improved quality of life and pain but **not survival**.
- **Current role:** docetaxel- and cabazitaxel-refractory disease, or marginal performance status.

## Platinum Agents

- **Not approved or widely used** in unselected mCRPC.
- **Platinum-sensitive subsets:** small cell/aggressive variant (cisplatin-etoposide or docetaxel-carboplatin; responses short); a randomised phase II showed **cabazitaxel + carboplatin** prolonged PFS in aggressive-variant but not usual mCRPC; DNA-repair-deficient tumours, e.g. **germline BRCA2** responding better to docetaxel-carboplatin than docetaxel alone.

### CHEMOTHERAPY AGENTS IN MCRPC: COMPARISON

| Agent              | Class / key property                                | Dose                                                          | Line                           | Survival benefit   | Signature toxicity                                                                 |
|--------------------|-----------------------------------------------------|---------------------------------------------------------------|--------------------------------|--------------------|------------------------------------------------------------------------------------|
| <b>Docetaxel</b>   | Taxane; microtubule stabiliser; blocks AR transport | 75 mg/m <sup>2</sup> IV 3-weekly + prednisone                 | <b>First-line</b> chemotherapy | Yes (TAX 327)      | Neutropenia, fatigue, oedema, <b>neuropathy</b> , hyperlacrimation, nail dystrophy |
| <b>Cabazitaxel</b> | Taxane; <b>poor P-glycoprotein affinity</b>         | 20 (or 25) mg/m <sup>2</sup> IV 3-weekly + prednisone + G-CSF | <b>Post-docetaxel</b>          | Yes (TROPIC, CARD) | <b>Neutropenia</b> , diarrhoea; little neuropathy                                  |

| Agent                          | Class / key property        | Dose                                          | Line                                   | Survival benefit            | Signature toxicity |
|--------------------------------|-----------------------------|-----------------------------------------------|----------------------------------------|-----------------------------|--------------------|
| <b>Mitoxantrone</b>            | Semisynthetic anthracycline | 12 mg/m <sup>2</sup> IV 3-weekly + prednisone | Palliative; taxane-refractory or frail | <b>No</b> (palliation only) | —                  |
| <b>Carboplatin / cisplatin</b> | Platinum                    | With etoposide or a taxane                    | Neuroendocrine; DNA-repair-deficient   | Subset-dependent            | —                  |

- **Pairs to hold:** docetaxel neuropathy vs cabazitaxel neutropenia/diarrhoea; mitoxantrone palliates but does not prolong life; cabazitaxel beats a second ARPI (CARD) but not docetaxel first line (FIRSTANA).

### CHEMOTHERAPY TRIALS IN MCRPC

| Trial            | Population                | Comparison                                   | Key result                                              | Verdict                                  |
|------------------|---------------------------|----------------------------------------------|---------------------------------------------------------|------------------------------------------|
| <b>TAX 327</b>   | Chemotherapy-naïve        | Docetaxel 3-weekly vs weekly vs mitoxantrone | OS 18.9 vs 17.3 vs 16.4 mo; HR 0.76                     | <b>3-weekly docetaxel = standard</b>     |
| <b>SWOG 9916</b> | Progressive CRPC          | Docetaxel + estramustine vs mitoxantrone     | OS 17.5 vs 15.6 mo; HR 0.80                             | Estramustine adds toxicity, not efficacy |
| <b>TROPIC</b>    | Post-docetaxel            | Cabazitaxel 25 vs mitoxantrone               | OS 15.1 vs 12.7 mo; HR 0.70                             | Cabazitaxel approved (2010)              |
| <b>PROSELICA</b> | Post-docetaxel            | Cabazitaxel 20 vs 25 mg/m <sup>2</sup>       | 20 non-inferior; fewer grade 3–4 events                 | <b>20 mg/m<sup>2</sup> default</b>       |
| <b>FIRSTANA</b>  | Chemotherapy-naïve        | Cabazitaxel 20/25 vs docetaxel 75            | No OS superiority                                       | Docetaxel stays first line               |
| <b>CARD</b>      | Post-docetaxel + one ARPI | Cabazitaxel vs alternative ARPI              | OS 13.6 vs 11.0 mo, HR 0.64; PFS 4.4 vs 2.7 mo, HR 0.52 | <b>Switch mechanism, not ARPI</b>        |

### Novel Chemotherapeutic Approaches

- **Antibody–drug conjugates (ADCs):** antibody to a cell-surface antigen delivers a cytotoxic payload — targets include PSMA, TROP-2, STEAP-1, tissue factor, delta-like protein 3 (DLL3).
- **Shared payload** (typically a microtubule inhibitor such as monomethyl auristatin E) → sensitivity depends on **target prevalence**; progression on one PSMA-targeted therapy (e.g. <sup>177</sup>Lu-PSMA) does not preclude response to a PSMA ADC.
- **Toxicity** stems from the payload and from target expression on normal tissue.

**Concept — taxanes are the only cytotoxics that extend life in mCRPC:** docetaxel first, cabazitaxel after it, with cabazitaxel's P-glycoprotein escape explaining why docetaxel failure does not predict cabazitaxel failure.

#### KEY POINTS: CYTOTOXIC CHEMOTHERAPY

- **Docetaxel is standard first-line chemotherapy:** prolongs PFS and OS, relieves pain, improves quality of life.
- **Docetaxel toxicity:** myelosuppression, fatigue, peripheral oedema, neurotoxicity, hyperlacrimation, nail dystrophy.
- **Cabazitaxel is second-line chemotherapy** after progression on or after docetaxel; 20 or 25 mg/m<sup>2</sup> 3-weekly both acceptable, the lower dose less toxic.
- **Cabazitaxel toxicity:** neutropenia (including febrile), diarrhoea; generally not neuropathy.

- **Docetaxel increasingly used earlier**; CRPC management depends on prior agents and response.
- **Mitoxantrone palliates without prolonging survival**; used after taxanes or when they would not be tolerated.
- **Platinum agents** for small cell prostate cancer and perhaps other subsets (combined TP53 and PTEN inactivation; DNA-repair-deficiency mutations).

## RADIOPHARMACEUTICALS

### Radium-223

- **Evolution**: beta emitters strontium-89 and samarium-153 (bone-pain palliation) → largely replaced by the **alpha emitter radium-223**.
- **Alpha physics**: path length **<100 µm** → spares adjacent marrow → limited haematological toxicity.
- **ALSYMPCA (landmark)**: symptomatic bone metastases, **no known visceral disease or bulky nodes**; 50 kBq/kg IV every 4 weeks × 6 → OS **HR 0.70**, median gain **3.6 months** (14.9 vs 11.3); time to first symptomatic skeletal event delayed ~6 months (HR 0.66); better quality of life (FACT-P).
- **Tolerability**: adverse and serious adverse events **fewer than placebo**; diarrhoea commonest; anaemia in about one-third (grade ≥3 in 13%).
- **Radium-223 + abiraterone/prednisone**: trial stopped early for **increased deaths and fractures** → label warning against the combination.
- **With any ARPI (e.g. enzalutamide)**: start or continue a **bone-protective agent** — zoledronic acid or denosumab (receptor activator of nuclear factor-κB ligand [RANKL] inhibitor).

#### ALSYMPCA: BODY TEXT VS FIG. 166.8

| Item     | Body text                   | Fig. 166.8               | Reason for difference                                                            |
|----------|-----------------------------|--------------------------|----------------------------------------------------------------------------------|
| Patients | 809                         | 921 (614 : 307)          | 809 = interim analysis; 921 = full randomised population in the updated analysis |
| OS       | HR 0.70; median gain 3.6 mo | 14.9 vs 11.3 mo; HR 0.70 | Consistent (updated analysis)                                                    |

### Lutetium-177 Vipivotide Tetraxetan PSMA Therapy

- **Theranostics has tempered radium-223 use** in favour of precision targeting.
- **PSMA**: transmembrane **carboxypeptidase** preferentially expressed on prostate cancer cells vs normal prostate and benign tissue → target for imaging and therapy.
- **FDA-approved PSMA PET tracers**: gallium-68 PSMA-11, piflufolastat F-18, flutufolastat F-18 — better detection before prostatectomy and in biochemical recurrence.
- **Phase II (post-chemotherapy, post-ARPI)**: PSA50 ~57%; objective response ~82% in measurable disease.
- **TheraP** (randomised phase II; PSMA PET-selected, chemotherapy-eligible): <sup>177</sup>Lu-PSMA-617 6.0–8.5 GBq 6-weekly ≤6 cycles vs cabazitaxel 20 mg/m<sup>2</sup> 3-weekly ≤10 cycles → **PSA response 65% vs 37%**, fewer grade 3–4 events, **no OS difference**.
- **VISION (landmark)**: after ≥1 ARPI and ≥1 taxane; 2:1 to <sup>177</sup>Lu-PSMA-617 + standard care vs standard care (chemotherapy not allowed as standard care) → imaging-based PFS **8.7 vs 3.4 months (HR 0.40)**; median OS **15.3 vs 11.3 months (HR 0.62)**; time to first symptomatic skeletal event 11.5 vs 6.8 months (HR 0.50).

- **VISION toxicity:** grade  $\geq 3$  events more frequent (52.7% vs 38.0%); commonest fatigue, **dry mouth**, nausea, anaemia.
- **FDA approval 2022:** mCRPC after ARPI and taxane. **PSMAfore** tests use before chemotherapy.

### New Radiopharmaceutical Approaches

- **Other PSMA beta emitters:**  $^{177}\text{Lu}$ -PSMA-I&T (phase III ECLIPSE).
- **Alpha emitters:** actinium-225-PSMA-J591 — possibly similar efficacy, less toxicity; alpha-beta combinations under study; optimal position in sequence unknown.

### RADIUM-223 VS LUTETIUM-177-PSMA-617

| Feature               | Radium-223                                                        | $^{177}\text{Lu}$ -PSMA-617                         |
|-----------------------|-------------------------------------------------------------------|-----------------------------------------------------|
| Emitter               | <b>Alpha</b> (path $< 100\ \mu\text{m}$ )                         | <b>Beta</b>                                         |
| Targeting             | <b>Bone-seeking</b>                                               | <b>PSMA-directed</b> (requires PSMA PET positivity) |
| Eligible disease      | Symptomatic <b>bone-only</b> ; no visceral or bulky nodal disease | PSMA-expressing metastases, any site                |
| Prior therapy (label) | Not required                                                      | <b>ARPI and taxane</b>                              |
| Schedule              | 50 kBq/kg IV 4-weekly $\times 6$                                  | 6-weekly, up to 6 cycles                            |
| Pivotal trial         | ALSYMPCA: OS HR 0.70                                              | VISION: OS HR 0.62, rPFS HR 0.40                    |
| Key toxicity          | Diarrhoea, anaemia; fewer AEs than placebo                        | Fatigue, dry mouth, nausea, anaemia                 |
| PSA                   | Rarely falls; flare possible                                      | Flare possible                                      |
| Combination caution   | <b>Never with abiraterone</b> ; bone agent with any ARPI          | —                                                   |

**Concept** — *radium follows bone, lutetium follows PSMA: the targeting mechanism defines eligibility — bone-only disease for radium, PSMA-avid disease at any site for lutetium.*

#### KEY POINTS: RADIOPHARMACEUTICALS

- **Radium-223 improves survival** in symptomatic bone-only mCRPC; use zoledronic acid or denosumab alongside, particularly with an ARPI.
- **$^{177}\text{Lu}$ -PSMA-617 improves survival** in PSMA-expressing mCRPC after ARPI and chemotherapy; use before chemotherapy is anticipated as data mature.

### POLY-ADP-RIBOSE POLYMERASE INHIBITORS AND DNA REPAIR TARGETING

- **PARP** repairs mainly **single-strand breaks**; cells lacking homologous recombination (the main double-strand repair route) depend on it.
- **Two mechanisms:** catalytic inhibition (competes with  $\text{NAD}^+$ ) and **PARP trapping** (cytotoxic PARP-DNA complexes; trapping potency differs between agents)  $\rightarrow$  replication-fork stalling  $\rightarrow$  double-strand breaks  $\rightarrow$  death without HRR = **synthetic lethality**.
- **Not effective in unselected patients;** targeted to somatic/germline HRR mutations — BRCA1/2, ATM, PALB2, FANCA and other Fanconi genes. Somatic DNA-repair mutations in mCRPC: up to **20–25%**.

- **AR-PARP synergy:** PARP1 also regulates transcription and promotes AR signalling; AR maintains HRR gene expression; ADT upregulates PARP-mediated repair → rationale for **ARPI + PARPI first line** (mirrors AR inhibition radiosensitising via impaired DNA repair).

### Poly-ADP-Ribose Polymerase Inhibitors as Single Agents

- **TOPARP-A** (olaparib, phase II; 50 heavily pretreated mCRPC): response ~33% overall but **~88% with DNA-repair mutations** (all BRCA2-loss patients had PSA falls >50%; 4 of 5 ATM); better rPFS and OS if biomarker-positive → **breakthrough designation** for BRCA1/2- or ATM-mutated mCRPC after taxane and ARPI.
- **TOPARP-B:** 98 patients with alterations in 18 DNA-repair genes; composite response 43/98; **BRCA1/2 best**, but activity in other HRR genes supported broader HRR use.
- **PROfound (landmark, phase III):** progression on a second-generation hormonal agent; olaparib vs enzalutamide/abiraterone; cohort A (BRCA1/2 or ATM): imaging PFS **7.4 vs 3.6 months (HR 0.34)**, OS 18.5 vs 15.1 months; benefit held across the 15-gene HRR panel → **FDA approval May 2020** for a broad HRR panel.
- **TRITON2** (rucaparib; 115 BRCA-altered, after ARPI and one taxane): objective response 43.5% (independent review) / 50.8% (investigator) in measurable disease; PSA response 54.8%; similar in germline and somatic and in BRCA1 and BRCA2, but **higher PSA response with BRCA2**.
- **TRITON3** (rucaparib vs physician's choice ARPI or docetaxel; 405 randomised after ARPI progression): imaging PFS **11.2 vs 6.4 months (HR 0.50)** in BRCA; 10.2 vs 6.4 (HR 0.61) intention-to-treat; **ATM no benefit** (8.1 vs 6.8, HR 0.95) → rucaparib approved for **BRCA1/2, not ATM**.
- **GALAHAD** (niraparib; single-arm phase II; BRCA1/2, ATM, FANCA, PALB2, CHEK2, BRIP1, HDAC2; after ARPI and taxane): median rPFS 8.2 months in the BRCA cohort → **breakthrough designation (2019)** for BRCA1/2-mutated mCRPC.
- **TALAPRO-1** (talazoparib; single-arm; DDR-HRR alterations; after 1–2 taxanes and ARPI): objective response ~30%; grade 3–4 anaemia commonest (31%).

### Poly-ADP-Ribose Polymerase Inhibitors in Combination With Androgen Receptor Pathway Inhibitors

- **PROpel** (phase III; first-line mCRPC after ADT): **olaparib + abiraterone** vs abiraterone; primary investigator-assessed rPFS; OS **42.1 vs 34.7 months (HR 0.81, P = 0.054)** — ~7-month gain; ~42 months is the **longest median OS in a first-line mCRPC phase III**; all subgroups favoured the combination, BRCA most → approved for BRCA-mutated mCRPC (2023).
- **MAGNITUDE:** **niraparib + abiraterone/prednisone** in HRR-altered mCRPC; primary rPFS → progression/death risk ↓ 45%, median rPFS >1.5 years; BRCA rPFS **19.5 vs 10.9 months (HR 0.55)**; also delayed symptomatic progression (HR 0.60) and chemotherapy (HR 0.67); **no benefit without HRR alteration** → approved for BRCA-mutated CRPC (2023).
- **TALAPRO-2:** **talazoparib + enzalutamide** vs enzalutamide first line (no prior ARPI; prior chemotherapy allowed); all-comers and HRR-deficient cohorts; benefit **regardless of biomarker**, greatest in BRCA1/2 → approved for HRR-mutated mCRPC (June 2023).

#### TALAPRO-2: BODY TEXT VS FIG. 166.10 (RADIOGRAPHIC PFS)

| Population | Talazoparib + enzalutamide | Placebo + enzalutamide | HR   | Source       |
|------------|----------------------------|------------------------|------|--------------|
| All-comers | Not reached                | 21.9 mo                | 0.63 | Fig. 166.10A |

| Population                                       | Talazoparib + enzalutamide | Placebo + enzalutamide | HR   | Source       |
|--------------------------------------------------|----------------------------|------------------------|------|--------------|
| HRR-deficient (subgroup of all-comers)           | 27.9 mo                    | 16.4 mo                | 0.46 | Fig. 166.10B |
| HRR non-deficient or unknown                     | Not reached                | 22.5 mo                | 0.70 | Fig. 166.10C |
| HRR-deficient (dedicated cohort, later analysis) | Not reached                | 13.8 mo                | 0.45 | Body text    |

- **Why the HRR-deficient values differ:** the figure is the HRR-deficient subgroup within the all-comers cohort; the body text reports the separate HRR-deficient cohort at a later analysis — different denominators and data cuts.

#### PARP INHIBITORS IN MCRPC: COMPARISON

| Agent              | Monotherapy evidence                    | Combination partner (trial)                 | Approved population                                    |
|--------------------|-----------------------------------------|---------------------------------------------|--------------------------------------------------------|
| <b>Olaparib</b>    | TOPARP-A/B; <b>PROfound</b> (phase III) | Abiraterone ( <b>PROpel</b> )               | Mono: broad HRR panel (2020); combination: <b>BRCA</b> |
| <b>Rucaparib</b>   | TRITON2; <b>TRITON3</b> (phase III)     | —                                           | <b>BRCA1/2 only (not ATM)</b>                          |
| <b>Niraparib</b>   | GALAHAD (single-arm)                    | Abiraterone/prednisone ( <b>MAGNITUDE</b> ) | Combination: <b>BRCA</b>                               |
| <b>Talazoparib</b> | TALAPRO-1 (single-arm)                  | Enzalutamide ( <b>TALAPRO-2</b> )           | Combination: <b>HRR-mutated</b>                        |

- **Pattern:** benefit is always largest in BRCA (especially BRCA2) and weakest or absent in ATM; only talazoparib + enzalutamide carries the broad HRR label among combinations.
- **Trap:** PROpel's OS result ( $P = 0.054$ ) was a trend, not a significant gain.

#### Biomarkers and End Points in Poly-ADP-Ribose Polymerase Inhibitor Trials

- **Response confined to HRR-deficient tumours;** panels vary (up to 15 genes vs limited sets). Evidence is clearest for **BRCA2**; weaker or inconsistent for others (e.g. ATM).
- **Early PARPI use appears advantageous;** trials extend into first-line mCRPC and mCSPC.
- **End points:** OS is the best yardstick but is diluted by effective subsequent therapy; **rPFS** = randomisation to first radiographic progression by blinded independent central review; **objective response rate (ORR)** = complete + partial response by RECIST, the commonest basis for FDA solid-tumour approvals; PSA and CTCs are less direct.

#### The Evolving Treatment Landscape of Poly-ADP-Ribose Polymerase Inhibitors and Side Effects

- **Hormone-sensitive trials:** **AMPLITUDE** (niraparib + abiraterone in HRR-mutated mCSPC); **TALAPRO-3** (TALAPRO-2 design in DNA-repair-altered mCSPC).
- **Class toxicity:** nausea, fatigue, **anaemia** (watch for sudden haemoglobin falls); rarely **myelodysplastic syndrome** and pulmonary embolism.
- **Switching PARPI** for intolerance is reasonable if eligible for more than one (e.g. olaparib or rucaparib in BRCA); toxicity matters more as use moves earlier.

**Concept — PARPIs are only as good as the repair defect they exploit:** BRCA2 loss gives true synthetic lethality, ATM loss does not — and ADT-induced repair dependence is why pairing with an ARPI works.

**KEY POINTS: PARP INHIBITORS**

- **Synergy with ADT:** ADT upregulates the DNA-damage-repair pathways PARP targets.
- **Several PARPis approved** as monotherapy and, recently, with abiraterone or enzalutamide for mCRPC.
- **Greatest benefit in specific HRR mutations**, especially BRCA1/2; optimal patient selection still under study.

**NOVEL MOLECULAR APPROACHES****Targeting ATR**

- **ATM-deficient tumours respond poorly to PARPis**, but **ATR is synthetically lethal with ATM** → ATR inhibitors may suit this subset.
- **Berzosertib:** randomised phase II, carboplatin–docetaxel vs carboplatin–berzosertib → stopped at interim for **futility** (fewer responses, more severe toxicity).
- **Ceralasertib (AZD6738):** tolerable with abiraterone (phase I); **TRAP** phase II with olaparib, with and without HRR deficiency (ATM loss or biallelic BRCA1/2 loss); primary end point response rate (RECIST or PSA fall ≥50%); eligibility after ≥1 mCRPC line.

**Targeting DNA Protein Kinase**

- **DNA-dependent protein kinase (DNA-PK):** part of **non-homologous end joining (NHEJ)**; drives progression via Wnt transcription; promotes glycolysis (ALDOA, PKM2 activity; pyruvate and lactate synthesis); explored as radio- and chemosensitiser; agents in development.

**Concept — each DNA-repair defect needs its own synthetic-lethal partner:** BRCA with PARP, ATM with ATR — a single "HRR-positive" label hides these differences.

**IMMUNOTHERAPY**

- **Why prostate cancer suits immunotherapy:** slow growth gives time to mount a response; tissue-specific antigens (PSA, **prostatic acid phosphatase [PAP]**).
- **Approaches tried:** dendritic-cell therapy, granulocyte-macrophage colony-stimulating factor (GM-CSF) adjuvant, viral vectors, whole-cell and single-antigen vaccines, DNA vaccines, cytotoxic T-lymphocyte antigen-4 (CTLA-4) and programmed cell death protein 1 (PD-1) blockade.
- **Only sipuleucel-T succeeded;** GVAX (allogeneic whole-cell), PROSTVAC-VF (poxviral PSA) and ipilimumab failed in late phase.

**Sipuleucel-T**

- **Autologous CD54+ dendritic cells** (collected by apheresis) loaded with a **PAP–GM-CSF fusion protein**.
- **Two phase II/III trials** missed their time-to-progression end points, but pooled analysis showed OS 23.2 vs 18.9 months (HR 0.67); OS was not primary, so no approval then; post hoc benefit seemed confined to Gleason ≤7.
- **IMPACT (landmark; FDA approval April 2010):** 512 patients, 2:1, asymptomatic or minimally symptomatic mCRPC; no visceral metastases or opioid use; 85% chemotherapy-naïve → **OS 25.8 vs 21.7 months (HR 0.78)** despite 64% placebo crossover.
- **No change in PFS or PSA/radiographic response** — an immunotherapy class effect (delayed action, flare); immune-related response criteria proposed.

- **Use:** approved for asymptomatic/minimally symptomatic CRPC; given mainly pre-chemotherapy; **avoid with visceral disease or opioid-requiring pain.** Toxicity mild (infusion fever, rigors).
- **Combinations** (ADT in biochemical recurrence, radiotherapy, ARPIs, radium-223, atezolizumab, ipilimumab): no proven clinical benefit.

### Immune Checkpoint Blockade

- **CTLA-4 blockade (ipilimumab):** phase I/II — PSA fall  $\geq 50\%$  in 10–20%, radiological response  $\sim 5\%$ ; immune-related events: colitis 15–20%, hepatitis, endocrinopathies incl. adrenal insufficiency, dermatitis/vitiligo, hypophysitis.
- **Post-docetaxel phase III** (799; 8 Gy to a bone metastasis then ipilimumab vs placebo): better PSA response and PFS, **OS not significant (HR 0.85, P = 0.053)**; best without visceral disease and with normal haemoglobin and alkaline phosphatase.
- **Pre-docetaxel phase III** (600): **no OS benefit (HR 1.11)** despite longer PFS.
- **Pembrolizumab:** KEYNOTE-028 response 17%; **KEYNOTE-199** (post-docetaxel and ARPI) response  $\sim 5\%$ .
- **Negative phase III combinations:** atezolizumab + enzalutamide; pembrolizumab + enzalutamide (KEYNOTE-641), + docetaxel (KEYNOTE-921), + olaparib (KEYLYNK-010) → **no role for checkpoint inhibitors in unselected mCRPC.**

### Pembrolizumab for Microsatellite Instability–High Cancers

- **Tumour-agnostic FDA approval (May 2017):** microsatellite instability-high (MSI-H) or mismatch repair-deficient (dMMR) solid tumours progressing on prior therapy; **200 mg IV 3-weekly**; response 40% across colorectal and non-colorectal cancers.
- **Prostate experience:** case-level complete, partial and near-complete PSA responses, concentrated in dMMR/MSI-H tumours.
- **Authors' recommendation:** pembrolizumab for MSI-H/dMMR mCRPC after  $\geq 1$  mCRPC line (e.g. abiraterone or docetaxel); also indicated for tumour mutational burden  $\geq 10$  mutations/megabase.
- **dMMR prevalence in mCRPC 2–5%;** test by next-generation sequencing or immunohistochemistry for MSH2, MSH6, MLH1, PMS2 → if positive, **genetic counselling for Lynch syndrome.**
- **Toxicity:** fatigue, pruritus, rash, gastrointestinal upset; immune-mediated colitis, hepatitis, endocrinopathy, pneumonitis, nephritis.

### Novel Immunotherapy Approaches

- **B7-H3:** T-cell co-regulatory molecule overexpressed in prostate cancer, linked with poor outcome; ADCs, monoclonal and bispecific antibodies and CAR-T in development. **Enoblituzumab** (neoadjuvant, localised disease): feasible, no surgical delay, grade 3 events in 12% (including one perimyocarditis).
- **Chimeric antigen receptor (CAR) T cells** — mostly PSMA-directed: early study tolerable with stable disease; TGF- $\beta$ -resistant PSMA CAR-T: 100% manufacturing success, **cytokine release syndrome** and one death, PSA falls  $\geq 30\%$  in 4 of 13; with low-dose IL-2: universal grade 3–4 toxicity, two partial responses. **Prostate stem cell antigen CAR-T:** dose-limiting grade 3 cystitis, manageable after modified lymphodepletion.
- **Efficacy not yet established;** toxicity potentially significant.

### Other Vaccine Strategies

- **Vaccine monotherapy is ineffective;** checkpoint upregulation causes resistance → combinations with checkpoint inhibitors.

- **Targets tried:** human telomerase reverse transcriptase, mixed peptide, DNA-PAP, PSA/brachyury/MUC-1, AR ligand-binding domain, personalised peptides; combined with ipilimumab, pembrolizumab, atezolizumab + sipuleucel-T — all under evaluation.

### IMMUNOTHERAPY TRIALS IN MCRPC

| Agent / trial                                                                         | Population                                                         | Result                                            | Verdict                |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------|------------------------|
| <b>Sipuleucel-T — IMPACT</b>                                                          | Asymptomatic/minimally symptomatic, no visceral disease or opioids | OS 25.8 vs 21.7 mo, HR 0.78; no PFS or PSA change | <b>Approved (2010)</b> |
| Ipilimumab (post-docetaxel)                                                           | After docetaxel + 8 Gy bone RT                                     | PFS better; OS HR 0.85, P = 0.053                 | Negative               |
| Ipilimumab (pre-docetaxel)                                                            | Asymptomatic/minimally symptomatic                                 | OS HR 1.11                                        | Negative               |
| Pembrolizumab — KEYNOTE-199                                                           | Post-docetaxel and ARPI                                            | Response ~5%                                      | Minimal activity       |
| Checkpoint combinations (atezolizumab + enzalutamide; KEYNOTE-641, -921; KEYLYNK-010) | mCRPC                                                              | No rPFS or OS gain                                | Negative               |
| <b>Pembrolizumab (MSI-H/dMMR)</b>                                                     | Tumour-agnostic                                                    | Response 40% across tumours                       | <b>Approved (2017)</b> |

**Concept — prostate cancer is immunologically cold except when it is hypermutated:** vaccines give survival without visible response, checkpoint blockade fails in unselected disease, and only the dMMR/MSI-H minority responds.

#### KEY POINTS: IMMUNOTHERAPY

- **Sipuleucel-T**, the first therapeutic cancer vaccine approved by the FDA, is indicated for asymptomatic or minimally symptomatic mCRPC **without** visceral metastases or opioid-requiring cancer pain.
- **Pembrolizumab** (PD-1 inhibitor) is indicated for MSI-H/dMMR mCRPC or tumour mutational burden  $\geq 10$  mutations/megabase — only 2–5% of prostate cancers.
- **Future immunotherapy:** novel targets, earlier use, combination with standard therapy.

## EMERGING THERAPEUTIC APPROACHES

### Bipolar Androgen Therapy

- **Bipolar androgen therapy (BAT):** depot testosterone while continuing LHRH agonist/antagonist → testosterone swings from **supraphysiological to near-castrate** over each 4-week cycle.
- **RESTORE** (phase II; testosterone cypionate 400 mg IM 4-weekly): PSA50 14–30% by cohort; after BAT, **enzalutamide rechallenge** gave PSA50 of 68% in previously enzalutamide-resistant men.
- **TRANSFORMER** (randomised phase II; post-abiraterone; BAT vs enzalutamide): primary PFS equal (5.7 months each; negative); post hoc **PFS2 28.2 vs 19.6 months** favouring BAT → enzalutamide (PSA50 to enzalutamide after BAT 77.8%).
- **Safety:** suitable only **without opioid-requiring pain or bulky visceral disease**; generalised pain, fatigue, peripheral oedema; improves energy, social function and sexual/erectile function.

## Prostate-Specific Membrane Antigen Targeting: Bispecific T-Cell Engager Approaches

- **Bispecific T-cell engagers (BiTEs)** bind CD3 on T cells and a tumour antigen (PSMA), following blinatumomab (CD19/CD3) in B-cell acute lymphoblastic leukaemia.
- **AMG 212, AMG 160, JNJ-63898081**: PSA responses, often transient, but **cytokine release syndrome** dose-limiting → inpatient monitoring and **tocilizumab** (anti-IL-6 receptor); development uncertain.

## PI3K/Akt/mTOR Inhibition

- **Pathway**: growth factor → receptor tyrosine kinase (or Ras/G-protein-coupled receptor) → PI3K converts PIP2 to PIP3 → Akt → mTOR (protein synthesis, cell growth); Akt blocks apoptosis via BAD, BAX, NF-κB, MDM2.
- **PTEN** (phosphatase and tensin homologue) dephosphorylates PIP3 → PIP2, terminating signalling; **PTEN loss in ~40%**; pathway altered in nearly half of primaries and virtually all metastases; **PIK3CA activating mutations rare**, PIK3R1/PIK3R3 loss-of-function common.
- **mTOR inhibitors failed** (rapamycin, everolimus, temsirolimus ± bicalutamide) — feedback upregulation upstream.
- **Reciprocal crosstalk**: PI3K inhibition activates AR in PTEN-deficient cells; enzalutamide activates Akt (via reduced PHLPP) → rationale for dual blockade; BEZ235 + abiraterone too toxic.
- **Ipatasertib** (pan-Akt inhibitor): phase II + abiraterone — rPFS benefit confined to PTEN loss; **IPATential150** (phase III, 1101 patients): rPFS in PTEN-deficient 18.5 vs 16.5 months (significant); intention-to-treat 19.2 vs 16.6 months (HR 0.84, not significant); grade ≥3 events 70% vs 39%; diarrhoea prominent. **Capivasertib + docetaxel** under study.

**Concept — blocking one growth pathway reactivates the other**: AR and PI3K signalling compensate for each other, so single-pathway inhibition gives modest gains and dual blockade is limited by toxicity.

## PALLIATIVE MANAGEMENT

### Pain and Spinal Cord Compression

- **Palliation and quality of life are the most important objectives**; cancer pain is the most debilitating symptom.
- **Focal bone pain → external-beam radiotherapy**; image painful scan-positive sites with plain radiographs or CT to exclude lytic lesions or pathological fracture, especially in limbs and weight-bearing bones.

**TABLE 166.1 COMMON PAIN SYNDROMES IN METASTATIC CASTRATION-RESISTANT PROSTATE CANCER**

| Pain syndrome                                 | Initial management                                                                                                                      | Other alternatives                                                                                                                                    |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Localised bone pain</b>                    | Analgesia; <b>localised radiotherapy</b> (weight-bearing, lytic, limb sites); assess for cord compression in <b>all focal back pain</b> | Surgical stabilisation of fracture or extensive erosion; radiopharmaceutical if local radiotherapy fails                                              |
| <b>Diffuse bone pain</b>                      | Analgesia; corticosteroids; bisphosphonate or RANKL inhibitor                                                                           | Multisite/wide-field radiotherapy; radiopharmaceuticals; calcitonin; chemotherapy                                                                     |
| <b>Epidural metastasis / cord compression</b> | <b>High-dose corticosteroids</b> ; analgesia; radiotherapy                                                                              | Physiotherapy; <b>surgical decompression and stabilisation</b> for high-grade compression, extensive bone involvement or post-radiotherapy recurrence |

| Pain syndrome                                                            | Initial management                                       | Other alternatives                                                              |
|--------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Nerve plexopathy</b> (tumour extension or prior therapy; rare)        | Analgesia; radiotherapy if not given before              | Tricyclic (amitriptyline); anticonvulsant (gabapentin, pregabalin); nerve block |
| <b>Other neurogenic</b> (post-herpetic neuralgia, peripheral neuropathy) | Full neurological assessment; analgesia                  | Tricyclic; anticonvulsant; <b>stop neurotoxic drugs</b> (docetaxel, platinum)   |
| <b>Uncommon</b> (skull base/cranial nerve, painful liver, pelvic mass)   | Radiotherapy; analgesia; corticosteroids (cranial nerve) | Chemotherapy; intrathecal chemotherapy for meningeal disease                    |

- **What the table is for:** radiotherapy is the backbone for localised pain; systemic and multisite measures for diffuse pain; steroids first for cord compression; neuropathic pain needs adjuvants (tricyclics, gabapentinoids), not just opioids.
- **Trap:** every focal back pain is cord compression until proven otherwise.

### Epidural cord compression

- **Common in prostate cancer** (vertebral and paravertebral spread); early treatment preserves ambulation and sphincter function.
- **Mostly from vertebral bodies;** less often paravertebral soft tissue — which may show only a **neurological deficit** with normal bone imaging.
- **Spinal MRI** is the routine investigation, replacing CT and conventional myelography.

**Sequence:** suspected compression → **high-dose IV dexamethasone** (16–100 mg/day; load 10 mg IV or 20–40 mg orally, then 4–10 mg every 6–12 h) → MRI → definitive radiotherapy ± decompressive surgery → taper steroids over 2–3 weeks once improved.

- **Decompressive surgery + radiotherapy beats radiotherapy alone** (Patchell randomised trial, 2005).
- **Operate if:** progression during radiotherapy; unstable pathological fracture; recurrence after radiotherapy — weighed against overall prognosis.
- **Chemotherapy rarely** used acutely.

### Bone-Targeted Approaches

- **Normal remodelling:** osteoclastic resorption → osteoblastic formation repairing the defect left by osteoclasts.
- **In prostate cancer:** ADT increases osteoclastic resorption; tumour resorbs matrix locally; cytokines, growth factors and bone morphogenetic proteins drive both cell types; metastases are **predominantly blastic**.
- **Hypercalcaemia is rare;** significant hypercalcaemia usually signals **neuroendocrine disease** (PTHrP).
- **Two uses:** ADT-related osteopenia/osteoporosis (oral bisphosphonates or RANKL inhibitor, per FRAX fracture risk) and **skeletal-related event (SRE) prevention** in metastatic disease (IV bisphosphonate or RANKL inhibitor) — fewer fractures, less pain, opioid use and admission, but **no antitumour effect**.
- **ARPIs and radium-223 also reduce SREs;** bone agents strongly recommended with radium-223; **radium-223 + abiraterone/prednisone not recommended.**

### Bisphosphonates

- **Mechanism:** inhibit osteoclast activity and proliferation.
- **Zoledronate:** reduced SREs vs placebo in progressive CRPC with bone metastases (randomised trial); zoledronate and pamidronate increase bone density on long-term ADT.

- **Dose:** 4 mg IV every 4 weeks, adjusted for creatinine clearance; supplement **calcium 1000 mg/day + vitamin D 800 IU/day**.
- **Toxicity:** fatigue, myalgia, fever, anaemia, creatinine rise, hypocalcaemia, **osteonecrosis of the jaw (ONJ)** — mostly with dental work or poor dentition → avoid in these patients.
- **Other bisphosphonates** (alendronate, etidronate, ibandronate, clodronate): benefit unproven.

### RANK Ligand Inhibitors

- **Mechanism:** tumour cytokines induce RANKL → activates RANK on osteoclasts; blocked by antibody or recombinant **osteoprotegerin** (natural decoy receptor).
- **Denosumab vs zoledronate (landmark phase III, 1904 bisphosphonate-naïve mCRPC):** time to first SRE **20.7 vs 17.1 months**; first and subsequent SREs HR 0.82; **no OS or PFS difference**.
- **Toxicity:** fatigue, nausea, hypophosphataemia, hypocalcaemia (grade  $\geq 3$  5%), **ONJ 2–4%** → calcium and vitamin D strongly encouraged.
- **Dose:** 120 mg SC every 4 weeks; **no renal adjustment** or monitoring.

### ZOLEDRONATE VS DENOSUMAB

| Feature         | Zoledronate                                                  | Denosumab                                          |
|-----------------|--------------------------------------------------------------|----------------------------------------------------|
| Class / target  | IV bisphosphonate; osteoclast inhibition                     | Fully human anti- <b>RANKL</b> monoclonal antibody |
| Dose            | 4 mg IV 4-weekly                                             | 120 mg SC 4-weekly                                 |
| Renal function  | <b>Dose-adjust</b> for creatinine clearance                  | <b>No adjustment</b>                               |
| SRE prevention  | Superior to placebo                                          | <b>Superior to zoledronate</b> (20.7 vs 17.1 mo)   |
| Survival effect | None                                                         | None                                               |
| Key toxicity    | Fatigue, myalgia, fever, creatinine rise, hypocalcaemia, ONJ | Hypocalcaemia, hypophosphataemia, ONJ 2–4%         |

### Other Palliative Needs

- **Hydronephrosis** from pelvic nodal progression: unilateral lymphoedema, rising creatinine, or incidental finding. **Not all need drainage** if the contralateral kidney functions; nephrostomies carry home-care burden; internal stents need regular change.
- **Circumoral numbness** → skull-base involvement → irradiate the skull base.
- **Paraneoplastic syndromes** are rare → consider neuroendocrine/aggressive variant.
- **Weight loss and anorexia:** usually treatment toxicity or progression — a normal sign that time may be limited; supplements can add distress; address family concern.

### Hospice and End-of-Life Care

- **Shift to symptom-directed care** when options are exhausted and performance declines (sedentary, housebound); discuss this phase as expected.
- **Hospice** adds support for pain and symptom control and improves bereavement; initiate in a timely fashion.

**Concept — bone agents protect the skeleton, not the patient's lifespan:** they reduce skeletal events without antitumour effect, whereas cord compression is the one palliative problem where hours, steroids and surgery change outcome.

**KEY POINTS: PALLIATIVE MANAGEMENT**

- **Back pain with known bone metastases → evaluate for cord compression:** back pain, focal motor/sensory deficit, or bladder/bowel change.
- **Immediate spinal MRI and high-dose corticosteroids;** definitive treatment is surgical decompression (if feasible), radiotherapy, or both.
- **Zoledronic acid and denosumab** both prevent SREs in castration-resistant bone metastases; denosumab needs no renal dosing; both can (rarely) cause ONJ.
- **Palliative and end-of-life care is multidisciplinary:** symptom control, clear communication of expected outcomes; hospice adds support.

**CONCLUSION**

- **Four open challenges:** (1) optimal sequencing of approved and experimental agents; (2) rational combinations exploiting feedback loops and alternative-pathway activation to overcome resistance; (3) smarter trials that discard failures quickly and fast-track winners; (4) patient selection by clinical and molecular features (DNA-repair mutations, PTEN loss).
- **Prospective biomarker-driven trials** — with tumour biopsies or correlative samples — are the only route to these answers.

**Concept** — *the problem has moved from finding drugs to ordering them: with many active agents, sequencing and biomarker selection now determine outcome.*